

Lecture 2 | Friday, 11 September 2026 | Lecturer : Yu Li

Section :

1.Data type

2.Pyton Programming

3. Sequence data and Sequencing Technologies

4.DNA pipeline and Sequence Allignment

Section 1: Data type

1.Sequential :

- ACTG sequence -> DNA
- ACUG sequence -> RNA
- ACTG, 5mC, 5hmC sequence -> epigenetic

What is 5mC , 5hmC ?

5mC : C + methylation -> 5-methylcytosine -> gene expression = **OFF**

5hmC : 5mC + oxidation -> 5-hydroxymethylcytosine -> gene activation = **ON**

2. data matrix : collection of records

- fixed attributes :
- n - object (people, gene etc)
- m = attribute (height, HbA1c etc)

3. Spatial : location , spatial information :

eg. location of a cell in an organ, protein structure , map

4. Temporal data : time data

eg. ECG, health data collected by wearables

5. Graph/ Network : show link between objects

eg. cell, protein interaction network , hub gene network

6. text : sentences / documents

eg. acedemic paper, social media, doctor words

7. multimodal : EHR, video etc

EHR (Electronic health records) : contain many type data above

eg. doctor writing (texts) , medical imaging (spatial) . WGS (sequential)

7. unknown : diet / exercise

reason for unknown : can be mixed or a single type above :

diet : data matrix (microbiome) , text (nutritionist recommendation)

exercise : heart rate from wearables (temporal) ,

Section 2 : Python programming

programming = communicate with computer -> ask it to do task

python = high level : more easy for human to understand

low level : more easy for machine to understand : eg. assembly , C++ etc

computer cannot understand human language

plug in package for python : remember **load them** before use it

Library :

NumPy: array processing

Pandas: dataframes & tables

SciPy: scientific stats

Matplotlib: base plotting

Seaborn: statistical plotting

Scikit-learn: classical machine learning

PyTorch: deep learning

Biopython: sequence parsing

Scanpy: single-cell RNA-seq

Squidpy: spatial transcriptomics

[] for array , eg [1,2,3,4,5] -> store 1,2,3,4,5 as value, a array with length 5

1. () for function , eg np.mean() , np.std() -> calculate the mean, standard deviation of an array

2.print() = output something

3.“ ” , ‘ ’ : word between them . eg. “The array is “

4.the comma sign : for separate different output

Example :

print(“ The array is “ , 3) :

output : The array is 3

print (“ This is “ , 1 , “ and this is “ , 2)

output : This is 1 and this is 2

Section 3 : Sequence data and Sequencing Technologies

Central Dogma : DNA -> RNA -> Protein

Type	DNA	RNA	Protein
category	A C T G	A C U G	20 amino acid
method	sequencing	sequencing	mass spectrometry

DNA -> RNA = **Transcription** , RNA -> Protein = **Translation**

Sequencing method :

Feature	1st Generation (Sanger)	2nd Generation (NGS)	3rd Generation (Nanopore)
Core Mechanism	Chain-termination via ddNTPs	Sequencing-by-synthesis (SBS) with reversible terminators	Single-molecule real-time sensing (electrical current / fluorescence)
Read Length	500 – 1,000 bp	100 – 300 bp (Short reads)	10 kb – >4 Mb (Ultra-long reads)
Throughput	Low (Single reaction)	Massive parallel / High-throughput	High
Raw Accuracy	Very High (>99.99%)	High (99.9%)	Moderate to High (99.0% – 99.75%)
Limitations	High cost per Mb, non-scalable	Cannot bridge complex repeats or large structural variants	Higher raw error rate, requires algorithmic error correction

Mass Spectrometry :

Step 1 : Enzymic digestion -> peptide fragment

Step 2 : HPLC/UV separation

Step 3 : Mass Spectrometer measures m/z (mass to charge ratios)

Step 4 : Algorithmic reconstruction -> full polypeptide sequence

Section 4 : DNA pipeline and Sequence Alignment

1. DNA processing pipeline :
 - output raw data in FASTQ
 - FASTQ = raw sequence data format
 2. Quality control : filtering -> remove low quality/unrelated data
 3. Mapping : read reference genome and alignment
 4. Variant calling : detect **single nucleotide variants** (SNVs)
- What is SNVs :
- Reference : ACCT**G**TC
- Sample : ACCT**C**TC
- G <-> C is a SNVs
5. Variant annotation : annotate **clinical** or **functional** impact

Alignment paradigm :
In lecture 3